Composio OAuth Compass

Autonomous Integration Launchpad, BYOC Architecture & MCP Server

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1. Direct Evaluation Deliverables

Live Production App:	https://composio-oauth-compass.vercel.app
GitHub Repository:	https://github.com/mrityunjay-tiwari/composio_oauth_compass
NPM Registry Package:	https://www.npmjs.com/package/composio-oauth-compass
MCP Global Command:	npx -y composio-oauth-compass
MCP Discovery Endpoint:	https://composio-oauth-compass.vercel.app/api/mcp

2. Executive Summary & Problem Understanding

Developers building AI agents on Composio experience a severe drop-off point when transitioning from Day-0 prototyping to enterprise production. While Composio Managed Auth enables 2-minute prototype setup, migrating to custom credentials (Bring Your Own Credentials / BYOC) exposes developers to a multi-week compliance cliff:

* The 7-Day Silent Expiry Trap: Google Cloud apps left in 'Testing' mode silently revoke refresh tokens after 7 days, causing agent cron jobs to fail every Sunday night.
* The $15,000 CASA Tier 2 Scope Penalty: Requesting broad scopes (e.g. gmail.modify instead of gmail.send) triggers mandatory third-party security audits costing $3,000 to $15,000 annually and 4 to 6 weeks in delays.
* The Slack App Directory Illusion: Developers mistakenly wait 14 days for App Directory approval when they only need to toggle Direct Public Distribution for instant 0-minute distribution.
* The Spotify Quota Lockout: Development mode is hard-capped at 25 whitelisted email accounts before cryptic 403 errors.
* Salesforce Cluster Replication Lag: Newly created consumer keys take 2 to 10 minutes to propagate across global clusters.

3. Product Pillars & System Architecture

Composio OAuth Compass solves these friction points through four tightly integrated systems:
1. Empirical Ground Truth Audit Matrix: Rigorously verified comparison between documented claims and tested developer reality.
2. Autonomous Live Investigation Studio: End-to-end agent probe that ingests raw documentation via Jina Reader, extracts OAuth parameters, and outputs ready-to-run Composio BYOC configs.
3. DevOps Boundary Matrix: Precise demarcation of responsibilities defining what Composio and agents automate versus where human DevOps must intervene (billing, DNS TXT, YouTube video screencasts, 2FA).
4. Model Context Protocol (MCP) Server: A globally distributed tool server on npm (composio-oauth-compass) enabling coding agents in Claude Desktop, Cursor, and Antigravity to query friction data directly.

4. Empirical Evidence & Ground Truth Matrix

Comparative analysis of official vendor documentation versus tested developer realities:

Provider	Official Doc Claim	Tested Developer Reality (Trap)	Compass Solution / Guidance
Google Cloud	Configure consent screen and beg...	7-Day Silent Expiry: In Testing mode, ...	Push to 'In Production' with internal/...
Google Scopes	Select requested scopes (e.g. gm...	$15,000 CASA Tier 2 Audit: Restricted ...	Downgrade to gmail.send (Sensitive) to...
Slack API	Submit app for review before dis...	App Directory Illusion: Multi-workspac...	Toggle 'Public Distribution' directly ...
Spotify API	Authenticate users with standard...	25-User Quota Gate: Development mode i...	Alerts developers to submit Quota Exte...
Salesforce	Create Connected App and connect...	10-Minute Cluster Lag: Newly created k...	Flags 10-minute replication delay so t...

5. Model Context Protocol (MCP) Server Specification

The MCP server is packaged and published globally on npm as composio-oauth-compass. It connects to Cursor, Claude Desktop, Antigravity IDE, and VS Code over standard stdio transport, exposing 6 specialized tools:

1. list_oauth_providers: Filter providers by domain category and maximum friction score.
2. get_oauth_guide: Full structured guide with human vs. agent steps, gotchas, and SDK snippets.
3. audit_scope_risk: Evaluates scopes and warns if CASA Tier 2 ($3k-$15k) audits will be triggered.
4. get_unwritten_gotchas: Returns unwritten platform failure modes omitted from official docs.
5. generate_composio_byoc_payload: Synthesizes valid Python/TypeScript Composio AuthConfig JSON.
6. investigate_oauth_provider: Probes unfamiliar platforms via Jina Reader and outputs BYOC guides.

6. Composio Bring Your Own Credentials (BYOC) Code Pattern

Every guide and MCP tool outputs ready-to-execute Composio SDK code:

```
from composio import Composio

composio = Composio()
# 1. Register Custom Auth Config (BYOC)
auth_config = composio.auth_configs.create(
    app="slack",
    credentials={"client_id": "CLIENT_ID", "client_secret": "CLIENT_SECRET"}
)
# 2. Generate Hosted Connect Link for User
connection = composio.connected_accounts.link(
    user_uuid="user_p7a89x",
    auth_config_id=auth_config.id
)
print("Direct user to authenticate:", connection.redirect_url)
```

7. Verification & Production Stack

Stack: Next.js 16 (App Router), Tailwind CSS v4, TypeScript 5, React 19, @modelcontextprotocol/sdk, esbuild.
Design System: Composio Cobalt Blue (#0024BC), sharp rounded-none geometry, official abcDiatype font, One Dark syntax-highlighted code blocks.
Test Suite: 100% pass rate across all 6 MCP tools via automated test runner (node mcp-test/test-all-tools.js).